



Day 10 of 60

Real lessons. Not textbook advice.

10 Things I Wish I Knew before starting ML.

🎓 For Beginners

⚙️ For Engineers

Nobody handed me these lessons. I earned them by building real projects, making real mistakes, and spending months undoing bad habits from tutorials.

📌 10 lessons

💡 Real stories

🕒 Both audiences

💬 Drop your lesson

→ **Swipe through all 10.** At least one of these will change how you think about ML.



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Day 10 / 60



01

Data beats algorithms — every single time.

I spent weeks fine-tuning a neural network. Then I cleaned my dataset properly. Accuracy jumped by 11% — overnight. The model didn't change. The data did.

Most beginners obsess over which algorithm to use. Engineers obsess over which data to trust. That mindset shift is everything.



Before switching models, always audit your data quality first. Bad labels beat bad algorithms every time.

🎓 Beginner: EDA before modeling

⚙️ Engineer: Invest in pipelines



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02

Simple models are not stupid models.

My first instinct was always: "More complexity = better performance." Wrong. A logistic regression model I built for a small classification task outperformed a deep neural net I spent 4 days tuning.

Complexity has a cost: training time, interpretability, compute, and maintenance. A model that works simply is a model that survives production.



Start with the simplest model possible. Beat it before upgrading. This is called the "baseline-first" mindset.

 **Beginner: Linear models first**

 **Engineer: Defend your baseline**



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03

80% of your time is data prep. Accept it.

I used to think ML was 80% modeling, 20% everything else. Nope. It's almost the exact reverse. Missing values, inconsistent labels, duplicates, distribution drift — you will fix all of this before writing a single `.fit()` call.

The engineers who embrace data prep are the ones who build models that actually work.



Treat data cleaning as a first-class engineering task, not a chore. Pandas + Great Expectations = power combo.

 **Beginner: Learn Pandas deeply**

 **Engineer: Validate pipelines**



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Day 10 / 60



04

Accuracy is a trap. Pick the right metric.

I once built a fraud detection model with 97% accuracy. I was proud. Then I realised the dataset had 97% non-fraud samples — so a model that predicts "not fraud" every time would score the same.

Accuracy means nothing on imbalanced datasets. Precision, Recall, F1, AUC-ROC — these are your real gauges.



Always check class distribution before choosing an evaluation metric. Never trust accuracy alone.

 **Beginner: Learn F1 & ROC-AUC**

 **Engineer: Business metric first**



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Day 10 / 60



05

Iteration speed is your superpower.

The best ML practitioners I've seen aren't the ones with the deepest math. They're the ones who can test a hypothesis, evaluate it, and move to the next one — fast.

Your first model will be wrong. Your tenth will be better. Your fiftieth might be great. Embracing fast cycles of "build → measure → learn" is what separates momentum from stagnation.



MLflow + lightweight experiment tracking from Day 1. You will thank yourself in Week 3.



Beginner: Ship weekly projects



Engineer: Track every run



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Day 10 / 60



06

A train/test split is not validation enough.

Early on, I would split my data once — 80/20 — and call it validation. Then I started seeing models that scored 89% on my holdout set but 71% on real-world data.

A single split is susceptible to luck. K-Fold cross-validation gives you a fair, averaged picture of true generalisation. Use it from the beginning.



`StratifiedKFold` for classification. `TimeSeriesSplit` for time-based data. Never ignore the split strategy.

🎓 Beginner: Use `cross_val_score`

⚙️ Engineer: Hold-out test set is sacred



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Day 10 / 60



07

A model in a notebook is not a product.

The hardest lesson I learned: getting 90% accuracy in a Jupyter notebook is the easy part. Deploying that model to serve 1,000 requests per second, monitoring for drift, handling edge cases, and keeping it updated — that's the real job.

Deployment is a completely different engineering discipline. Start learning it early.



Flask/FastAPI → Docker → a cloud endpoint. That is the MVP deployment stack every ML engineer should know.

 **Beginner: Deploy one small API**

 **Engineer: Own your deployment**



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Day 10 / 60



08

Understanding your data is a skill, not a step.

EDA — Exploratory Data Analysis — used to feel like a checkbox to me. "Plot a few histograms, check correlations, done." But the more projects I did, the more I realised: every interesting model finding was first discovered visually, during EDA.

Your data tells a story. Your job is to read it before you start modelling.



Ask: What is the shape? Where are the outliers? Is there leakage? Does this distribution make sense? Before. Every. Model.

 **Beginner: Seaborn pairplots**

 **Engineer: ydata-profiling**



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Day 10 / 60



09

Track every experiment or you will repeat your mistakes.

I once spent 3 days re-discovering a hyperparameter configuration I had already tested and discarded a week earlier — because I didn't log it properly. Never again.

Experiment tracking isn't optional. It's how you build on your past self instead of fighting it. Your future self will be furious if you skip this.



MLflow, Weights & Biases, or even a simple CSV log. Something is infinitely better than nothing.

 Beginner: Log params + metrics

 Engineer: MLflow from Day 1



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Day 10 / 60



10

Building in public accelerates learning 10x.

I was afraid to share my work publicly for months. "It's not good enough." "People will judge." But the day I started posting my projects, experiments, and lessons — the feedback, connections, and opportunities I received completely changed my trajectory.

Your imperfect project shared today beats a perfect project hidden forever.



Push every project to GitHub. Write one LinkedIn post about what you learned. The compounding effect is real.



Beginner: Post your first project



Engineer: Write, share, teach



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Day 10 / 60

💬 What would you add?

These 10 lessons took me months to learn the hard way.

Drop your #11 in the comments below.

Let's build this list together. 🙌

Today's 10 at a glance:

- | | |
|-------------------------------|---------------------------------|
| 📁 01 — Data beats algorithms | 🔑 06 — Cross-validation matters |
| 🔪 02 — Simple models first | 📦 07 — Notebooks ≠ products |
| 🔪 03 — 80% is data prep | 🔍 08 — EDA is a real skill |
| 📉 04 — Accuracy is a trap | 💾 09 — Track experiments always |
| 🔄 05 — Iteration is your edge | 👥 10 — Build in public |

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Day 10 / 60